

Reasoning in Signals: A Survey of Large Language Models for Sensor Data Understanding

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Abstract—The application of Large Language Models (LLMs) to sensor data analysis is an emerging and impactful direction of research, which combines natural language reasoning with time-series and physical-world signals. This survey provides a comprehensive review of recent advances in applying LLMs to understand, reason about, and generate sensor data across domains such as health monitoring and Internet of Things (IoT). We categorize existing work by their input modalities, including raw signals, textual summaries, and visual prompts, as well as their application domains and technical strategies such as prompt engineering, sensor-text alignment, and hybrid frameworks. Despite encouraging results in low-dimensional and API-supported tasks, LLMs still struggle with complex signal processing and numerical reasoning, as well as multimodal sensor fusion. We also identify key challenges such as limited dataset availability, inadequate signal transformation, and the need for architectural adaptations. Future research directions are outlined to bridge these gaps. Our review emphasizes the potential of LLMs to serve as foundation models for analyzing sensor data, while also pointing out the importance of domain-specific knowledge and diverse modalities.

Index Terms—large language models, sensor data, multimodal AI

I. INTRODUCTION

The ability to interpret and reason over sensor data is foundational to build a wide range of intelligent systems, from wearable health monitors to smart homes and autonomous vehicles [1], [2]. Traditionally, sensor processing relies on domain-specific models and signal processing expertise [3]–[5], which creates a barrier to accessibility and scalability. Recently, large language models (LLMs) [6]–[9], initially designed for natural language tasks, have demonstrated reasoning capabilities on various types of data. Their abilities beyond textual domains have sparked growing interest in areas such as sensor data understanding.

This survey systematically reviews the current landscape of LLMs applied to sensor data across general, health, and Internet of Things (IoT) domains. We observe two trends in these works: efforts to align non-textual sensor signals with the text-native architectures of LLMs, and hybrid systems that combine LLM reasoning with specialized processing tools or APIs. Notable contributions include benchmarks for evaluating LLM performance on signal processing tasks; novel encodings

of sensor data into textual, visual, or semantically aligned forms; and integration of LLMs into interactive systems for diagnostics, monitoring, and control. Figure 1 presents a conceptual mind map of the paper’s organization, while Table I provides a structured overview of the works reviewed.

Despite rapid progress, current approaches face several limitations. LLMs often excel in simple and structured scenarios but underperform on complex tasks requiring domain reasoning or high-dimensional input handling. The lack of standard datasets and sensor-aligned pretraining further constrains their utility in real-world settings. Our review organizes the literature by data modality and task type, synthesizes the core techniques and findings, and identifies challenges and open questions that need to be addressed to realize LLMs as general-purpose sensor reasoning engines. The remainder of this paper is organized as follows: Section II reviews recent work on applying LLMs to sensor data understanding, covering general sensors, IoT applications, and healthcare contexts. Section III analyzes key challenges and limitations identified across these studies. Finally, Section IV outlines future research directions and concludes the paper.

II. LLMs FOR INTERPRETING SENSOR DATA

A. General Sensor Data

The application of LLMs to general sensor data tasks has primarily focused on benchmarking, model adaptation, and prompt strategies for structured time-series understanding. Quan et al. [10] introduce SensorBench, a comprehensive benchmark designed to systematically evaluate the capabilities of large language models in sensor data processing tasks. Motivated by the growing use of LLMs as “copilots” for engineering applications, the authors investigate whether LLMs can effectively support or potentially replace human experts in signal processing, particularly within cyber-physical and IoT systems. Their key contributions include a SensorBench Benchmark that have 11 tasks spanning 9 sensor modalities including ECG, audio, gait, pressure, with tasks categorized across four difficulty levels based on compositionality and parameterization. The benchmark includes 250 real-world and synthetic challenges, such as filtering, imputation, echo cancellation, and change point detection. LLMs including GPT-4o,

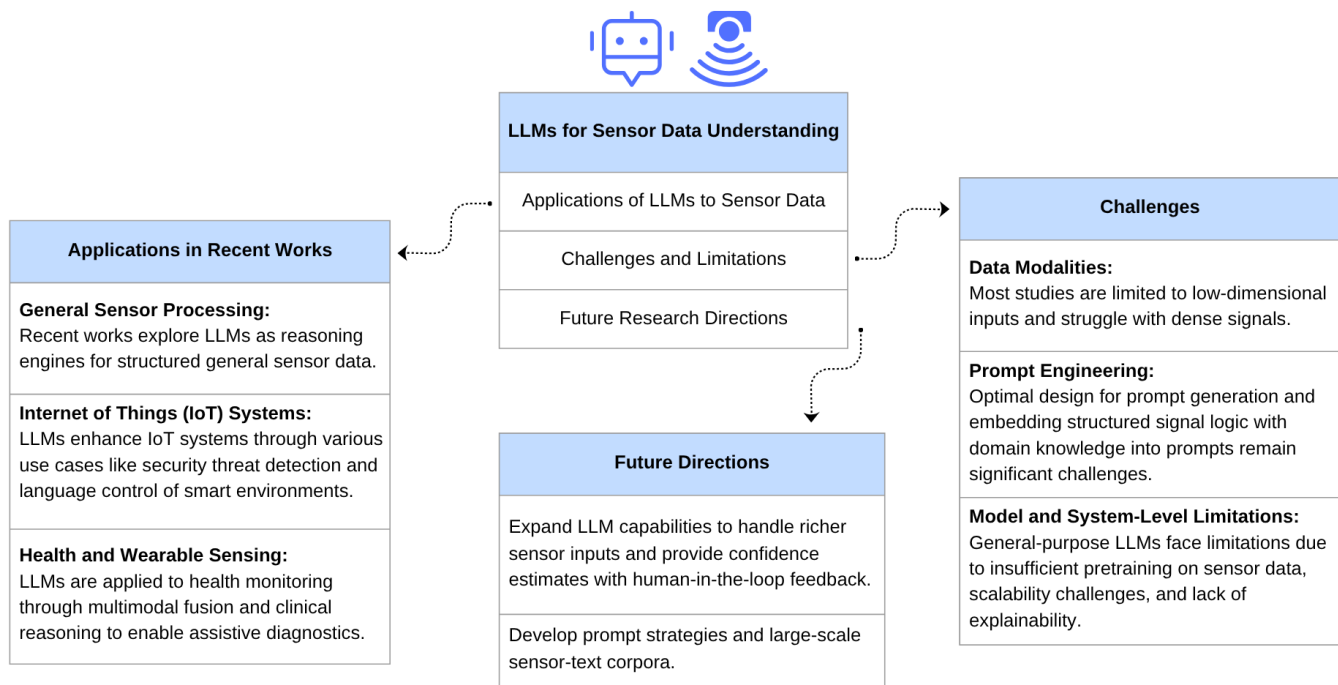


Fig. 1. A conceptual mind map summarizing key components of this survey.

GPT-4, GPT-3.5-turbo, and LLaMA-3-70B have been evaluated on the benchmark under three interaction modes: coding with APIs, coding without APIs, and text-based reasoning. The mode of coding with APIs yields the best overall performance. While LLMs perform comparably to experts on simple tasks such as resampling and delay detection, they significantly underperform on complex tasks including spectral filtering of speech, change point detection, heart rate estimation. In 64% of tasks, human experts outperform LLMs, which shows limitations of LLMs in domain-specific reasoning and parameter tuning. The study also reports several additional findings: prompting strategies, particularly self-verification, yield the best results. Fine-tuning on sensor-processing corpora shows minimal improvement for larger models such as LLaMA-3-70B, likely due to saturation, but yields more notable gains for smaller models like LLaMA-3B. The study also shows that LLMs tend to retrieve surface-level knowledge rather than perform deeper reasoning or planning, especially on compositional or parameter-sensitive tasks. SensorBench represents the first structured benchmark for assessing LLMs in signal processing, and the results indicate that while LLMs show promise for simple API-supported scenarios, they continue to struggle with complex reasoning tasks compared to trained human engineers. Overall, self-verification prompting currently outperforms fine-tuning in enhancing LLM performance in this domain. Closing the performance gap will likely require architectural advances in LLM reasoning and planning, rather than merely more data or narrow task-specific training.

Okita et al. [11] propose SENvT, a self-supervised transformer-based large language model architecture specifi-

cally designed for processing 1D sensor data, with targeted applications in activity recognition and emotion detection. Drawing inspiration from the remarkable success of LLMs in natural language processing and computer vision, the authors aim to establish a foundation model for the sensor signal domain using unsupervised learning techniques. This study explores multiple pretext tasks and finds that contrastive learning and masked token prediction are the most effective strategies for representation learning. SENvT trained with contrastive learning outperforms ResNet-based self-supervised learning models in transfer learning scenarios, although it underperforms during fine-tuning, likely due to the increased parameter complexity of transformer architectures. For masked token prediction, the model achieves optimal performance with a 60% masking rate during fine-tuning and 40% masking during transfer learning, suggesting that task-dependent masking strategies are critical to success. Experiments with multivariate SENvT models show slightly reduced performance, which the authors attribute to limited sensor diversity in the dataset with only three channels. Overall, this work highlights the emerging potential of transformer-based LLMs in time-series sensor learning, and supports the broader vision of developing foundation models for health-related AI and wearable computing applications. It also points to several promising directions for future research, including cross-device sensor fusion, multimodal integration, and addressing the computational scalability limitations of Transformer architectures when applied to long or high-resolution sensor streams.

LLMs have been established as foundation models for sensor-rich environments. Xu et al. [12] introduce the concept

TABLE I
OVERVIEW OF REVIEWED SYSTEMS FOR LLM-BASED SENSOR DATA UNDERSTANDING

Proposed System	Objective	Data Used	Methodology
SensorBench [10]	Sensor signal processing	ECG, audio, gait, pressure	Evaluates LLMs in API coding, non-API coding, and text reasoning modes.
SEnVt [11]	Analyze 1D sensor data	Emotion and activity recognition datasets with 1D multivariate signals	Builds a transformer model trained via contrastive learning and masked token prediction; tested under transfer and fine-tuning settings.
Penetrative AI [12]	Reason over textualized and digitized sensor data	Summarized signals and raw time-series	Prompts LLMs with structured sensor summaries and raw data; two interaction modes explored with LLMs.
SensorLLM [13]	Analyze human activity sensor data	Accelerometer and gyroscope data for HAR tasks	Two-stage framework: sensor-to-text alignment using Chronos encoder, and task-aware tuning with frozen LLM parameters.
Zhou et al. [14]	Sensor data synthesis	Synthetic sensor data across multiple domains	Task decomposition with LLMs; subtask delegation to DGMs; includes reinforcement learning for quality control.
Jia et al. [15]	Natural language interaction with building sensors	Real-time building sensor database	Predefined LLM prompts and API function integration for sensor queries; evaluated through case study and professional feedback.
Zong et al. [16]	Use LLMs in IoT domains	Network traffic, multi-device code bases, large-scale IoT sensor streams	Three case studies: DDoS detection, macroprogramming, and high-level sensor trend analysis using GPT-4.
LLMind [17]	Use LLMs in IoT domains	IoT device control data in smart environments	FSM-based prompt-to-code pipeline; memory-augmented planning and semantic retrieval from historical scripts.
IoT-LLM [18]	Use LLMs in IoT domains	Human activity recognition, anomaly detection, and localization data	Three-stage pipeline: data simplification, knowledge retrieval, structured prompting.
HealthLLM [19]	Health diagnosis	Four public datasets from wearable sensors: mental health, sleep, activity, metabolic signals	Evaluated 12 LLMs on 10 tasks; HealthAlpaca fine-tuned model with structured contextual enhancement.
By My Eyes [20]	Use MLLMs for sensor-based tasks	Accelerometer, ECG, EMG, respiration signals visualized as images	Converts sensor sequences into spectrograms and waveforms; auto-selects best visual format; paired with textual prompts.
DrHouse [21]	Health diagnosis	Wearable sensor data and medical texts	Multi-turn LLM prompts combined with sensor queries and probabilistic reasoning using decision-tree guidelines and adaptive data filters.

of Penetrative AI, where large language models such as ChatGPT are used not only for processing text but also for directly interpreting and reasoning about real-world physical phenomena through sensor data. Unlike traditional applications of LLMs limited to natural language, Penetrative AI treats LLMs as general-purpose reasoning engines embedded within cyber-physical systems, that are capable of understanding both textualized and digitized sensor signals. Their study explores two primary interaction modes: textualized signals, where human-readable summaries of sensor data such as step counts, WiFi signal strength, or satellite readings are provided as prompts to the LLM, and digitized signals, where raw numerical time-series data such as ECG waveforms are embedded directly within the prompt as structured inputs. For textualized signals, the authors demonstrate the feasibility of using ChatGPT-4 for user activity recognition, such as determining motion states and contextual awareness. With the addition of expert-designed prompts and chain-of-thought reasoning examples, ChatGPT-4 achieves up to 100% accuracy in motion detection and over 90% accuracy in environmental classification tasks, showing its strong generalization capability when appropriately guided. For digitized signal processing, the paper

investigates whether LLMs can understand and interpret raw sensor outputs directly. In one experiment, ECG waveform data are included in the prompts, and ChatGPT-4 is tasked with identifying R-peaks and estimating heart rate. Surprisingly, in certain configurations, the LLM performs comparably or better than the classical Pan-Tompkins algorithm in terms of heart rate estimation accuracy, showing not only signal comprehension but also robustness to noise and variation when using well-crafted prompts. The authors conclude that LLMs, when supplied with clear task framing and domain-relevant reasoning patterns, can outperform traditional signal processing methods in select contexts, particularly when flexibility and interpretability are needed. This work opens new directions for embedding LLMs into CPS and IoT applications, where they can expand the capabilities of intelligent sensing systems. The results also prove the importance of prompt engineering and multimodal task formulation for LLMs to reach their full potential.

Li et al. [13] propose SensorLLM, a novel two-stage framework that enables large language models to interpret and classify human activity from sensor data such as accelerometer and gyroscope readings, a domain typically dominated by special-

ized time-series models. The central challenge addressed is the inherent mismatch between the semantic, text-based structure of LLMs and the numerical, temporal nature of sensor data. SensorLLM bridges this gap through a two-stage process: the sensor-language alignment stage and the task-aware tuning stage. In the sensor-language alignment stage, the system automatically converts multi-channel sensor signals into trend-descriptive natural language using a combination of predefined templates, domain-specific tokens, and a time-series encoder. This stage produces textual representations of sensor trends without requiring manual annotation, effectively aligning the raw numerical input with a form that is natively interpretable by LLMs. By transforming time-series data into structured text, this step makes it possible for general-purpose LLMs to reason over sensor inputs using their existing language-based capabilities. In the task-aware tuning stage, SensorLLM applies these aligned textual descriptions to perform human activity recognition. Importantly, the models' parameters are kept frozen, preserving its general knowledge, while a lightweight classifier and projection module are trained for the downstream task. This approach makes efficient adaptation to time-series tasks possible without the need for large-scale fine-tuning. SensorLLM is the first framework to align raw sensor data with LLMs at a human-interpretable, semantic level, marking a significant step toward the development of multimodal foundation models for sensor-based reasoning. This framework demonstrates that LLMs combined with task-specific alignment techniques can be extended beyond language into domains like wearable computing, mobile health monitoring, and zero-shot time-series classification. It shows that there are new possibilities for using general-purpose LLMs in real-world sensor applications.

To generate synthetic data for sensor domains, Zhou et al. [14] propose a novel framework that integrates the reasoning capabilities of large language models with the generative precision of domain-specific generative models (DGMs). The innovation here is to use LLMs to understand and analyze data generation tasks, decompose complex tasks into modular subtasks, and delegate each subtask to the most suitable DGM for execution. This results in automated and adaptive data generation pipelines that can be customized to a wide range of sensor domains, and makes flexible and scalable synthetic data creation possible. This framework also introduces the potential of incorporating reinforcement learning to optimize the delegation process, further improving both the quality and control of the generated outputs. By using LLMs to learn how to strategically assign subtasks to different DGMs based on performance feedback, the system can achieve more efficient and targeted data synthesis. Experimental results show this collaborative approach between LLM and DGM significantly outperforms traditional generative methods, particularly in terms of generalization to diverse sensor tasks.

Jia et al. [15] present a case study exploring a protocol that uses a large language model to perform natural language interaction with sensor data in the context of building management. The primary goal is to empower users, particularly

those without technical backgrounds, to access and interpret real-time building sensor data through plain language queries, eliminating the need for specialized tools or programming skills. The system integrates with a database of building performance metrics and employs a set of predefined prompts alongside API-based functions to automate common tasks such as data retrieval and signal processing. Preliminary evaluations suggest that this approach can significantly improve the accessibility and usability of digital twin systems, and can also make advanced building analytics more approachable for a broader range of stakeholders. Feedback from professionals in construction and education sectors confirms its adaptability and user-friendliness, but also identifies key areas for further improvement, such as higher domain-specific accuracy and more command coverage. This case study demonstrates the feasibility of embedding LLMs into building information systems to support interactive, human-centric interfaces, and points a promising direction for future work as AI capabilities and sensor infrastructure continue to evolve.

These studies collectively demonstrate that LLMs are capable of supporting and, in some cases, outperforming traditional approaches in general sensor understanding, particularly when guided by structured prompts or paired with domain-specific encoders.

B. IoT Domain

In the IoT domain, LLMs have shown strong potential for orchestrating complex system behavior and enhancing automation. Zong et al. [16] investigate the potential of large language models, specifically GPT-4, to improve the functionality and usability of IoT systems through a series of case studies. The first case study looks at the role of LLMs in cybersecurity, demonstrating how GPT-4 can be used to detect Distributed Denial of Service (DDoS) attacks by analyzing network traffic patterns and providing interpretive insights that go beyond conventional rule-based methods. The second case study explores macroprogramming, where LLMs are used to automatically generate, adapt, and manage code across multiple IoT devices, thereby simplifying development workflows and reducing the technical burden on programmers, particularly in large distributed environments. The third case study examines sensor data processing, showing that LLMs are able to assist in advanced analysis and interpretation of large-scale sensor datasets. This includes tasks such as anomaly detection, contextual reasoning, and summarization of sensor trends in natural language.

To address the need for intelligent task coordination in IoT systems, Cui et al. [17] propose LLMind, an AI framework that integrates large language models with domain-specific AI modules and IoT devices to execute complicated real-world tasks through natural language interaction. In this framework, the LLM functions as a centralized orchestrator, responsible for interpreting user intent, planning multi-step procedures, and coordinating the actions of various computational modules and physical devices. Users engage with the system via familiar communication platforms such as WeChat, and issue

natural language commands that are automatically translated into executable control logic. At the core of the proposed LLMind is a finite-state machine-based language to code transformation pipeline, which serves as a bridge between user input and system execution. This pipeline performs task decomposition, diving it into modular components, supports robust code generation, and allows for fine-grained control of device behaviors. The proposed framework also incorporates semantic retrieval mechanisms that draw from a repository of historical scripts in order to enhance system efficiency by avoiding redundant computation. In addition, LLMind uses a combination of short-term and long-term memory coordination, so that the system can adapt over time through continuous user interaction and feedback. The effectiveness of LLMind is demonstrated through real-world scenarios, including automated guest check-in via facial recognition, and dynamic WiFi bandwidth allocation based on context-aware prioritization. These use cases prove the framework's ability to translate complex multi-agent tasks into reliable system actions. The authors position LLMind as a step toward achieving Artificial Complex Intelligence (ACI), which is an intermediate form of intelligence that lies between narrow, task-specific AI and fully general-purpose AI by allowing the system to evolve its capabilities through iterative refinement and human-in-the-loop guidance.

An et al. [18] propose IoT-LLM, a unified framework that uses large language models to perform reasoning over IoT sensor tasks by integrating data preprocessing, retrieval-augmented generation, and prompt engineering techniques. The framework is designed to bridge the gap between the raw, numerical nature of IoT sensor data and the semantic reasoning capabilities of LLMs. It consists of three main stages: an IoT data simplification and enrichment stage, a knowledge retrieval stage, and a prompt configuration stage. In the first stage, raw IoT sensor streams are first downsampled, tokenized with digit-level spacing, and augmented with metadata such as measurement units and device context. In addition, statistical features including mean, standard deviation, and FFT are extracted to support interpretability and enhance task relevance. These preprocessing steps make sure that the resulting input is both compatible with LLM input constraints and rich in contextual information at the same time. The second stage constructs two vector databases: a domain knowledge base and a demonstration base. A hybrid retrieval strategy combining keyword search and embedding-based similarity with re-ranking is used to automatically fetch relevant external knowledge and example tasks. This allows the system to scale to new applications without requiring handcrafted expert demonstrations or rules. The last stage uses role-based prompting and chain-of-thought reasoning templates to guide internal inference. All elements, including simplified sensor data, retrieved knowledge, metadata, and structured prompt templates, are merged into a unified prompt, so that LLMs can process and reason over sensor-based inputs effectively. To evaluate this proposed framework, the authors introduce a new benchmark of five real-world IoT tasks including human

activity recognition, industrial anomaly detection, heartbeat anomaly detection, human sensing, and indoor localization. The framework is tested using several LLMs, including GPT-3.5, GPT-4o-mini, Claude-3.5, Gemini-Pro, Mistral-7B, and LLaMA2-7B. Results show that the proposed IoT-LLM substantially boosts performance across all tasks. For example, GPT-4o-mini achieves an average performance gain of 49.4%. In complex cases like ECG anomaly detection, the model is able to generate expert-level insights. Ablation studies confirm that each of the three stages contributes to the overall improvement.

Together, these works demonstrate the evolving role of LLMs in making IoT systems more programmable and intelligent, which can be achieved when sensor data is preprocessed and when applying structured prompting techniques.

C. Health Wearable Sensors

Applications of LLMs in health-related sensing focus on multimodal fusion, and contribute to clinically relevant decision support using wearable data. Kim et al. [19] adapt large language models to consumer health prediction tasks using multimodal inputs from wearable sensor data and contextual information such as user demographics and health knowledge. Their study systematically evaluates 12 state-of-the-art general-purpose LLMs like GPT-3.5 and GPT-4, as well as domain-adapted biomedical models, across 10 prediction tasks spanning mental health, sleep, physical activity, and metabolic assessment, using four public datasets. The authors also introduce HealthAlpaca, a fine-tuned version of MedAlpaca, which achieves state-of-the-art performance in 8 out of 10 tasks, outperforming larger models through targeted domain adaptation. One of the key findings is on the effectiveness of context enhancement strategies. By incorporating structured knowledge such as user profiles, medical background, and time-series features, performance gains up to 23.8% can be obtained. Additionally, this study explores various prompting strategies, showing that while zero-shot prompting yields surprisingly competitive results, few-shot prompting and chain-of-thought reasoning significantly enhance prediction accuracy, particularly for larger and more capable models.

Yoon et al. [20] propose a novel approach for using multimodal large language models (MLLMs) to solve sensor-based tasks by converting raw time-series data into visual inputs, rather than relying on traditional text-based prompts. The motivation stems from the limitations of textual input for representing long, high-frequency numerical sequences, which often leads to inefficiencies and poor performance. To address this challenge, the authors introduce a visual prompting paradigm in which sensor data is visualized as images and paired with textual task instructions, allowing MLLMs to process complex sensor inputs through their vision-language interface. A key innovation in the proposed approach is the introduction of an automatic visualization generator, which uses in-context learning to select the most suitable visualization format for each task and data type using publicly available tools. Rather than relying on manual heuristics or

domain expertise, the generator evaluates alternative visualization strategies, such as choosing between waveform and spectrogram, and selects the most informative one based on visual output to make the approach adaptable. The proposed framework is evaluated across nine diverse sensory tasks spanning four modalities: accelerometer, ECG, EMG, and respiration. Experimental results show that visual prompting consistently outperforms text-only prompting, with an average accuracy gain of 10% and a 15.8 times reduction in token cost. These improvements are particularly significant for tasks involving long or high-frequency time series, where token limitations in LLMs often pose a bottleneck. Experiments also prove the effectiveness of the visualization generator, as inappropriate or fixed visualizations can lead to substantial drops in model performance. Moreover, the authors suggest that visual prompting remains robust even in low-shot scenarios compared to purely text-based approaches. This work marks an important step toward extending the utility of foundation models to multimodal reasoning in real-world environments.

To improve diagnostic accuracy and interactivity in virtual healthcare systems, Yang et al. [21] address critical limitations in existing LLM-based virtual doctor systems, which often rely heavily on patients' subjective symptom descriptions that can be vague, inconsistent, or inaccurate, and can cause frequent misdiagnoses. Moreover, current models typically lack access to real-time physiological data, due to outdated or static medical knowledge bases, and are confined to single-turn question-answer formats that fall short of replicating the interactive nature of real clinical consultations. To overcome these challenges, the authors propose DrHouse, a multi-turn, LLM-powered virtual doctor framework that integrates sensor data from wearable devices, continuously updated medical knowledge from authoritative sources such as UpToDate and PubMed, and formal diagnosis guidelines structured as decision trees. The proposed framework provides back-and-forth consultations that iteratively refine the diagnostic process, and introduces probabilistic reasoning by outputting likelihood estimates for multiple possible conditions to improve interpretability and clinical relevance. It actively combines real-time sensor data, patient self reports, and expert guidelines to perform context-aware diagnostic reasoning. A key feature here is its adaptive data retrieval mechanism, which uses a BERT-based semantic filter to determine when additional sensor data is needed, optimizing computational cost and latency. It also incorporates uncertainty handling by evaluating sensor data quality and, when necessary, recommending follow-up in-lab tests. The system architecture includes a vectorized knowledge base constructed from medical textbooks, diagnostic dialogues, and clinical guidelines; an indexing of sensor data with metadata and uncertainty measures; a multi-source retrieval using both mapping-based and semantic search techniques; structured prompt generation through fusion of user input, domain knowledge, and diagnostic rules; and probabilistic inference that integrates symptom-demographic priors with guideline-based likelihoods. Experimental results show that DrHouse outperforms baseline models in differenti-

ating between symptomatically similar diseases, for example, gastritis and hyperthyroidism, by using objective physiological data such as heart rate. Furthermore, the model demonstrates resilience in the face of noisy or incomplete sensor data, suggesting clinical testing when necessary. The paper shows that integration of dynamic knowledge sources, real-time physiological inputs, and multi-turn reasoning can provide a significantly more accurate and explainable alternative to conventional virtual diagnostic tools.

These studies show that when LLMs are combined with real-time sensor inputs and structured domain knowledge, they can be applied as powerful assistive healthcare applications.

III. CHALLENGES AND LIMITATIONS

Despite progress in this field, using large language models for sensor data understanding remains a complex research topic. Existing approaches have shown promising directions but have also exposed underlying limitations such as data modality, prompt formulation, scalability, interpretability, and system-level integration. This section analyzes some of the challenges identified across recent studies.

A. Data Modalities and Representations

A significant portion of current work has focused primarily on low-dimensional, 1D time-series data such as accelerometer signals, ECG, and simple IoT telemetry [10], [18]. While this choice reduces complexity and facilitates prompt engineering, it excludes high-dimensional modalities that are common in many real-world sensing applications. For example, SensorBench [10] evaluates LLM performance exclusively on 1D signals and does not consider visual or quantized representations, which may be more appropriate for complex or high-frequency inputs. Similarly, IoT-LLM [18] currently handles only low-dimensional signals and lacks support for multi-sensor fusion or spatial-temporal data, limiting its generalizability to complex environments.

Recent advances in multimodal LLMs have opened up visual prompting as an alternative to textual encoding of sensor data. However, Yoon et al. [20] find that visual prompting struggles with tasks requiring fine-grained numerical reasoning or dense multi-channel inputs, such as EEG with hundreds of channels. In some cases, increasing the number of examples or shots does not lead to better performance, which contradicts with assumptions from few-shot learning in text domains. Additionally, selecting an appropriate visualization for each task remains difficult, as fixed or poorly chosen formats can degrade model performance.

B. Prompt Engineering

The process of transforming raw sensor signals into text-based prompts that are interpretable by LLMs remains a non-trivial challenge. Current strategies vary from handcrafted summaries to automated feature extraction, but optimal approaches for different data types remain unclear. Penetrative AI [12] suggests that there is a need to better understand how physical concepts can be encoded and reasoned about

by LLMs. This includes designing prompts that embed domain knowledge, support reasoning chains, and capture signal structure effectively. Moreover, CoT-style prompting, which has shown success in traditional NLP tasks, does not consistently improve performance in sensor reasoning scenarios [20]. This suggests that reasoning over quantitative or structured physical data may require fundamentally different prompting strategies.

C. Model Limitations

Current general-purpose LLMs often fall short in reasoning about sensor data due to limited exposure to physical-world tasks during pretraining. Penetrative AI [12] identifies clear boundaries in what these models can conceptually understand, particularly in relation to physics-grounded tasks. While fine-tuning and alignment using sensor-text datasets is suggested as a solution, such resources remain scarce. Furthermore, studies such as [19] and [21] point to ethical, clinical, and practical challenges when deploying LLMs in sensitive domains like healthcare. These include concerns over privacy, demographic bias, and trustworthiness of probabilistic outputs, particularly when models operate with unbalanced or noisy inputs.

D. System-Level Challenges and Interpretability

From a systems perspective, LLM-based sensor reasoning frameworks face challenges in scalability and real-time responsiveness. For example, DrHouse [21] reports unstable latency from API calls that limit its utility in real-time clinical settings. Future work may explore edge deployment or integration with lightweight LLMs to mitigate this issue. Similarly, [12] and [15] suggest that meaningful integration of LLMs with procedural tools (e.g., code interpreters, control systems) remains underdeveloped. Expanding LLM frameworks to support multi-turn, interactive, and event-driven workflows remains an open area for both research and engineering.

Several works raise concerns about how LLM outputs align with expert expectations and mental models. For instance, medical professionals evaluating DrHouse noted that its diagnostic reasoning style differed from that of human clinicians [21]. This points to a need for explainable reasoning and human-in-the-loop workflows. Addressing these gaps will require interdisciplinary efforts involving sensor signal processing, machine learning, and domain-specific expertise.

IV. FUTURE DIRECTIONS AND CONCLUSION

Large language models have shown significant potential for reasoning over sensor data across a wide array of applications. This survey has reviewed recent advances that adapt LLMs for interpreting structured, time-series, and multimodal sensor inputs. This review summarizes a growing body of work that applies these models to tasks ranging from heart rate estimation to building automation, using diverse approaches such as sensor-text alignment and visual prompting. Yet, substantial challenges remain. These include limited generalization to high-dimensional data, difficulties of translating numerical patterns into semantically rich prompts, heavy reliance on

handcrafted and ad hoc workflows, and ethical concerns in high-stakes domains like healthcare.

That said, several important avenues remain open for future exploration. Future work should investigate how LLMs and multimodal foundation models can process richer data modalities, including video, audio, and multi-sensor fusion, particularly in environments with spatial-temporal complexity. Combining LLMs with modality-specific encoders or cross-modal transformers could support more generalizable reasoning. There is also a growing need for systematic approaches to design prompts that are tailored to sensor reasoning tasks. Research should explore automated prompt generation and hybrid strategies. Combining domain knowledge and structured metadata may improve reasoning over continuous and noisy signals. Moreover, current LLMs are pretrained on language-rich corpora but lack exposure to physical-world data and tasks. Building large-scale, curated corpora of sensor-text pairs or multimodal logs could improve their performance in physical-world domains. As LLMs take on more critical roles in domains like healthcare, it becomes essential to design interfaces that support transparency and interpretability. Future frameworks should provide mechanisms for confidence estimation and user feedback integration. This is particularly important when LLMs outputs differ from expert reasoning styles or face uncertainty in probabilistic tasks.

By continuing to refine how LLMs engage with sensor data, in both reasoning and interaction, researchers can achieve more intelligent systems that are not only powerful but also interpretable, context-aware, and aligned with human needs. These systems will be capable of understanding complex physical environments, adapting to multimodal inputs, and making informed decisions as assistive AI agents.

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